CHEM 405.LM1: Chemistry Principles for Engineers Laboratory
Fall 2021

Instructor: Gordy Hunter
Contact: Best method is using the "conversations" feature on myCourses (the inbox icon on the blue navigation bar on the left side of the screen)
Lab Meetings: Thursdays, 6:01 – 9:00PM
Laboratory: 88 Commercial St., Room 574

Important note:
This is an in-person laboratory section. There are no "online versions" of experiments available. If you miss an experiment, you will receive a score of zero for that lab. One lab grade will be dropped at the end of the semester.

Required materials:
- Mask that covers nose and mouth
- Safety glasses or goggles
- Lab coat

Course Description
CHEM 405 lab is a supplement to the CHEM 405 lecture and counts as 20% of your total grade in the course. Please note that you must pass the lab in order to pass the course. There are no lab make-ups, except in the event of a school cancellation. Two unexcused absences from lab will result in a failing grade for CHEM 405.

You are expected to be in lab on time and prepared to start that day's experiment. This means that you have read the background and procedure for the experiment and have completed the experiment's pre-lab questions. This also means that you will enter the lab wearing appropriate clothing, close-toed shoes, and a mask.

Handouts of the procedures and pre-lab questions will be provided in lab. Copies of these handouts will also be available to view on myCourses. You will also need to record and submit your data and observations for each experiment, but you will not need a dedicated laboratory notebook. A scientific calculator is also highly recommended for ease of calculations during the lab.

Pre-lab questions must be attempted and completed <u>before</u> each lab period and will be turned in <u>at the beginning of the lab period</u>, before the prelab discussion begins. In-lab observations and the "Data and Calculations" pages along with any post-lab questions are to be completed immediately after the experiment and submitted before leaving the lab.

Appropriate clothing must be worn in the laboratory. Lab coats, goggles, and a mask must also be worn while performing all experiments. These items can be purchased through any online site or retailer you choose, such as Work'n'Gear on South Willow St. in Manchester. <u>"Community" goggles are not available</u>, and a face shield is not an appropriate substitute for safety glasses. Your shoes must have closed toes and cover the top of the foot; no sandals or flip-flops are allowed. Suitable clothing should be worn underneath lab coats; shorts, capris and short skirts are inappropriate for lab work. If you are not dressed appropriately for lab, you will not be allowed to perform the experiment, and you will receive a zero for that experiment.

Grading
Labs will be graded on a 20-point scale:
6 points – pre-lab questions
6 points – in-lab observations
8 points – "Data and Calculations" pages and post-lab questions

Schedule of experiments

Sept 2	Introduction to the Chemistry Lab
Sept 9	Lab #1. The Atomic spectrum of hydrogen
Sept 16	Lab #2. Resolution of Matter I: Paper chromatography
Sept 23	Lab #3. The Geometrical Structure of Molecules
Sept 30	Lab #4. Resolution of Matter II: Fractional crystallization
October 7	Lab #5. Determination of an unknown chemical formula
October 14	**Midterm exam** (meet in the laboratory)
October 21	Lab #6. Molar mass from freezing point depression
October 28	*No lab this week*
November 4	Lab #7. Heat effects and calorimetry
November 11	Lab #8. Spectroscopy – Using light as a probe
November 18	Lab #9. Rates of chemical reactions: a clock reaction

Note: No labs after November 18